# 📅 Unit 2 Homework — Exponents & Scientific Notation

Complete independently · Exponent Rules · Scientific Notation

## A. Exponent Rules — Product & Quotient

**Problem 1**

Simplify: $x^5 \cdot x^6$

answer

**Problem 2**

Simplify: $b^9 \div b^4$

answer

**Problem 3**

Simplify: $(4x^2)(2x^5)$

answer

**Problem 4**

Simplify: $24m^8 \div 6m^3$

answer

**Problem 5**

Simplify: $c^3 \cdot c^4 \cdot c^2$

answer

**Problem 6**

Simplify: $(7y^4)(2y)$

answer

## B. Power, Zero & Negative Exponents

**Problem 1**

Simplify: $(y^5)^2$

answer

**Problem 2**

Simplify: $(3x^2)^2$

answer

**Problem 3**

Simplify: $12^0$

answer

**Problem 4**

Simplify: $(8m^3n^2)^0$

answer

**Problem 5**

Simplify: $2^{-3}$ (write as a fraction)

answer

**Problem 6**

Simplify: $y^{-2}$ (write as a fraction)

answer

**Problem 7**

Evaluate $-2^4$ (no parentheses).

answer

## C. Scientific Notation — Conversions

**Problem 1**

Write 81,000 in scientific notation.

answer

**Problem 2**

Write 0.00056 in scientific notation.

answer

**Problem 3**

Write $9.4 \times 10^5$ in standard form.

answer

**Problem 4**

Write $6.02 \times 10^{-3}$ in standard form.

answer

**Problem 5**

Write 12,300,000 in scientific notation.

answer

## D. Scientific Notation — Operations

**Problem 1**

Multiply: $(4 \times 10^2) \times (2 \times 10^6)$

answer

**Problem 2**

Divide: $(9 \times 10^8) \div (3 \times 10^2)$

answer

**Problem 3**

Multiply: $(5 \times 10^3) \times (6 \times 10^4)$ — adjust if needed!

answer

## E. Word Problem

**Problem 1**

Light travels $1.86 \times 10^5$ miles per second. How far does it travel in 10 seconds? (scientific notation)

answer